Parameter-Controlled Regime Transitions in Energy-Flow Cosmology:

Cross-Sector Consistency Between Lensing and Background Observables

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Abstract

Energy-Flow Cosmology (EFC) carries pre-registered observational predictions in two sectors: P1 specifies a sign-changing crossover in the effective weak-lensing coupling $\Sigma_{\text{eff}}(z)$ at $z \approx 0.44$ (DOI 10.6084/m9.figshare.32037990); P2 specifies a sealed linear growth prediction $f\sigma_8(z = 0.7) = 0.430$ (DOI 10.6084/m9.figshare.32013156). Both seals live in the sealed EFC codebase but occupy disjoint parameter spaces, and no unified forward-model mapping between them has been constructed. This work reports direct scans of the sealed lensing engine (`lensing.py`) that identify the kinetic stiffness scale K_0 as the primary control parameter governing the lensing-sector crossover redshift ($\Delta z \approx 0.87$ across $K_0 \in [0.5, 4.0]$), while amplitude parameters contribute only $\Delta z \approx 0.11$ across a $\times 50$ rescaling range. Cross-validation with a BBKS transfer function confirms the sign-change prediction is robust under power-spectrum choice, with quantitative shifts bounded below $\Delta z = 0.06$. Against the 13-framework landscape addressing Φ/Ψ-slip in the Symbiose Framework Atlas (2026-04-21 snapshot), EFC is the only framework in the surveyed set predicting a non-monotonic sign change; all other frameworks in the surveyed set predict monotonic profiles, a structural distinction we document in Table 1. Independently, both sectors predict $\mu < 1$ (entropic damping of the Newtonian potential) despite operating in disjoint parameter spaces, providing qualitative cross-sector consistency not shared by ΛCDM, DGP, Galileon, or $f(R)$ among the surveyed frameworks. We pre-register a symmetric falsification framework built on these two discrimination vectors (D1, D4) with two auxiliary vectors (D2, D3) reported as conditional secondary tests. Stage-IV data from Euclid DR1 (October 2026) and DESI DR2 will provide the joint falsification dataset. The phenomenological-to-action mapping (a required open task) is documented explicitly as limitation.

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1 Motivation

Energy-Flow Cosmology (EFC) predicts a regime transition between linear and non-linear dynamics in the late universe, motivated by the action defined in DOI 10.6084/m9.figshare.31876324 and developed through Technical Notes I (DOI 31333414) and II (DOI 31333600). The mechanism is tested via pre-registered predictions: P1 specifies a sign-changing crossover in the effective weak-lensing coupling $\Sigma_{\rm eff}(z)$ at $z \approx 0.44 \pm 0.03$ (DOI 32037990), to be observed by Euclid DR1 tomography (October 2026); P2 specifies a sealed linear growth prediction $f\sigma_8(z = 0.7) = 0.430 \pm 0.02$ (DOI 32013156), consistent with DESI DR2 within current precision.

A cross-reference against the 42-framework cosmological Atlas (Symbiose Research, snapshot hash recorded in §7) reveals that EFCs P1 occupies a structurally distinctive position within the surveyed set. We emphasize explicitly that the comparative claims in this work refer to the set of frameworks analysed in this paper and cross-referenced in the Atlas snapshot 2026-04-21; claims of the form "only framework" are scoped to this surveyed set and should not be interpreted as absolute uniqueness across all possible models of gravity. Among 13 frameworks in the Atlas that address the Φ/Ψ-slip phenomenon — EFC, Verlinde 2016, Horndeski, Galileon, $f(R)$_Starobinsky, DGP, MOND, EDE, Multi-Field-EDE, Quintessence, Thawing-Quintessence, w0waCDM, and ΛCDM — **EFC is the only framework in the surveyed set predicting a non-monotonic sign-changing $\Sigma_{\rm eff}(z)$ profile**. Every other framework in the surveyed set predicts a monotonic profile. This structural distinction, documented in Table 1 (§2) and confirmed against the primary literature (Peirone et al. 2018, Phys. Rev. D 97, 043519 for Horndeski; Pogosian & Silvestri 2016, arXiv:1606.05339 for the underlying conjecture), is the central empirical position of this work.

Direct inspection of the public EFC codebase (commit hash §7) reveals that P1 and P2 live in disjoint parameter spaces. P2 (Tech Note II) parameterizes the transition through (z_t, n, μ_0), determining a background gate function $\Theta(z)$ and a perturbation-level growth modification $\mu(a) < 1$. P1 (the lensing engine, `lensing.py`) parameterizes the same postulated mechanism through action-level quantities $(K_0, V', \alpha, \rho_{\rm crit}, \gamma_0)$, deriving (μ, η, Σ) from V'-driven λ-dynamics versus Ricci-sourced backreaction. Here K_0 acts as the effective kinetic stiffness scale in the EFC action, controlling the relative balance between V'-driven dynamics and Ricci-sourced backreaction. No explicit functional mapping $(z_t, n, \mu_0) \to K_0$ exists in the current implementation, and the two frameworks yield different values of $\mu(z)$ at fixed z ($\sim$11% difference at $z = 0.44$). This reflects the current operational structure of the framework: **consistent sign predictions under independent parameterizations**, not a unified forward-model. The action-to-phenomenology mapping required for strict identifiability is flagged as an open task (§6.1).

The pre-registered question addressed in this work is therefore bounded:

> *Does the sealed EFC lensing engine produce a sign-changing $\Sigma_{\rm eff}(z)$ profile under wide parameter variation, and does the sign of its sector-level μ-prediction agree with the sign of the P2 growth-sector μ-prediction despite disjoint parameterizations?*

To characterize the degree to which the lensing-sector crossover position is dynamically determined rather than freely tunable, we perform two controlled scans on the sealed lensing engine. First, a ×50 joint rescaling of amplitude parameters (α, V', γ_0) at fixed $K_0 = 1.37$ produces only $\Delta z \approx 0.11$ in crossover position, with the sign change persisting in every tested configuration. Second, a $\times 8$ scan of K_0 at fixed amplitude produces $\Delta z \approx 0.87$, identifying K_0 as the dominant control. Cross-validation with a full BBKS transfer function (§3.2.1) confirms the sign-change prediction is robust under power-spectrum choice; quantitative shifts in crossover position are bounded below $\Delta z = 0.06$.

On this empirical basis, we pre-register a primary falsification test based on two discrimination vectors — D1 (non-monotonic lensing crossover) and D4 (cross-sector $\mu < 1$ sign consistency, decomposed into D4a growth-sector sign and D4b combined consistency, §2.2) — and two conditional auxiliary vectors — D2 (Horndeski-forbidden μ-Σ quadrant) and D3 (preregistered $E_G = 1.086$, pending kSZ-reconstructed estimators). The four-vector framework is specified in §2; the test protocol in §5.

2 Discrimination Vectors

This section catalogues two primary discrimination vectors and two auxiliary (conditional) vectors against which Stage-IV data can test EFC. The distinction between primary and auxiliary reflects which vectors are directly observable with currently accessible or imminently released data.

2.1 D1 (primary): Non-monotonic lensing slip

EFC prediction: Sign-changing $\Sigma_{\text{eff}}(z)$ crossover at $z = 0.44 \pm 0.03$, preregistered L2, KILL badge (DOI 32037990). Engine derivation from V'-vs-Ricci balance in the action (DOI 31876324); polynomial calibration at Euclid DR1 bin centres in the sealed P1 module.

Comparison landscape: Of 13 frameworks in the Atlas snapshot (2026-04-21) addressing Φ/Ψ-slip, EFC is the only one in the surveyed set predicting non-monotonic sign change. Horndeski theories in particular have been shown by Peirone et al. (2018) to satisfy $(\Sigma - 1)(\mu - 1) \geq 0$ across statistically-independent Monte-Carlo ensembles, confirming the Pogosian & Silvestri (2016) conjecture. See Table 1.

Test data: Euclid DR1 cosmic-shear tomography (release October 2026).

Falsifier: Monotonic $\Sigma_{\text{eff}}(z)$ with no sign change in $0.3 \lesssim z \lesssim 0.55$ at $\geq 3\sigma$ significance.

2.2 D4 (primary): Cross-sector $\mu < 1$ sign consistency

EFC prediction: Both the sealed lensing engine and Tech Note II produce $\mu < 1$ at $z \approx 0.44$ despite operating in disjoint parameter spaces:

- Lensing sector (P1 engine at fiducial $K_0 = 1.37$, $k = 0.05\ h/\text{Mpc}$): $\mu = 0.823$
- Growth sector (P2 Tech Note II at $z_t = 1.01$, $n = 6$, $\mu_0 = 0.913$): $\mu = 0.922$

Both values lie below unity. Quantitative difference: $\Delta\mu \approx 0.099$ ($\sim$11%). We emphasize that this is a **sign-level consistency** prediction, not a quantitative identity.

Comparison landscape: Atlas framework predictions for $f\sigma_8(z = 0.7)$ within the surveyed set: EFC 0.430 ($\mu < 1$), ΛCDM 0.452 ($\mu = 1$), DGP 0.465 ($\mu > 1$), Galileon 0.48 ($\mu > 1$). $f(R)$_Starobinsky also predicts $\mu > 1$ via chameleon mechanism. Within the surveyed set, EFC is unique in predicting $\mu < 1$ in the growth sector among frameworks with quantified $f\sigma_8$ predictions.

Test data: DESI DR2 RSD $f\sigma_8(z)\times$ Euclid DR1 $\Sigma_{\text{eff}}(z)$.

Decomposed falsifiers. D4 is logically decomposed into two sub-falsifiers to allow independent interpretation:

- **D4a (growth-sector sign):** DESI DR2 RSD measurement of $f\sigma_8(z = 0.7)$ consistent with ΛCDM (0.452) or above ($\mu \geq 1$ regime) at $\geq 3\sigma$. This falsifies the growth-sector $\mu < 1$ component independently of D1.
- **D4b (combined cross-sector consistency):** Combined condition of D4a failure AND D1 failure (no lensing sign change). This falsifies the shared-mechanism hypothesis as a joint structural claim, beyond either individual sector failure.

The distinction matters for interpretation: D4a failure alone falsifies P2 (growth-sector prediction) but leaves P1 potentially viable; D4b failure rejects the cross-sector consistency hypothesis that motivates this paper.

2.3 D2 (auxiliary): Horndeski-forbidden μ-Σ quadrant (P5)

EFC prediction: $\mu = 0.85 \pm 0.1$, $\Sigma = 1.18 \pm 0.08$ at $z \approx 0.44$ and cosmological scales (preregistered L2, KILL badge).

Comparison: Pogosian & Silvestri (2016, arXiv:1606.05339) conjectured $(\Sigma - 1)(\mu - 1) \geq 0$ in viable Horndeski theories. Peirone et al. (2018, Phys. Rev. D 97, 043519) confirmed this conjecture across Monte-Carlo ensembles of statistically-independent viable Horndeski models, stating explicitly: *"(μ today < 1, Σ today > 1) quadrant is entirely absent from the predictions of the stable theory."*

Status: D2 is listed as auxiliary because the full (μ, Σ) posterior requires likelihood-pipeline MCMC (Cobaya + MGCAMB) on DES Y6 or Euclid DR1, which is specified in the companion test paper (`EFC_RCMP_Test_Specification_v1.1`) and has not yet executed.

Falsifier (if test executes): (μ, Σ) posterior excluding $(\mu < 1, \Sigma > 1)$ quadrant at $\geq 3\sigma$.

2.4 D3 (auxiliary): Preregistered $E_G = 1.086$ (P3)

EFC prediction: $E_G^{\mathrm{EFC}}/E_G^{\mathrm{GR}} = 1.086 \pm 0.012$, scale-independent in quasi-static regime (preregistered L2, KILL badge).

Comparison: Among the 42 Atlas frameworks in the 2026-04-21 snapshot, only EFC carries a preregistered quantitative E_G prediction within the surveyed set. Other frameworks in the surveyed set either adopt the GR baseline (1.0) or have no specific prediction in the Atlas.

Critical caveat: Existing RSD-β-based E_G measurements (e.g., Wenzl et al. 2024, ACT DR6 $\times$ BOSS giving $E_G(0.555) = 0.31$) inherit GR-Kaiser RSD templates that break under modified-gravity forward evaluation (Hu & Raveri 2025). D3 is therefore **not currently testable** with existing estimators; a valid test requires kSZ-reconstructed velocity estimators (Kamran et al. 2026, arXiv:2510.27605, Phys. Rev. D 113, 023527). We list D3 as an auxiliary future-validation vector.

Status: Pending kSZ-reconstructed E_G data from ACT + SO $\times$ Euclid galaxy cross-correlations. Likely earliest test date: 2027–2028.

Falsifier (when test becomes available): kSZ E_G/E_G^{GR} inconsistent with 1.086 ± 0.03 at $\geq 3\sigma$.

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Table 1: 13-framework Φ/Ψ-slip landscape

Source: Symbiose Framework Atlas phenomenon query `phi_psi_slip`, snapshot 2026-04-21 (hash §7). Primary literature confirmation for Horndeski column: Peirone et al. (2018).

Framework	Lens	Slip profile	Sealed
EFC	entropic_emergent	**Non-monotonic at** $z{\approx}0.44$	**Yes**
EDE	dyn._dark_energy	Monotonic	—
Multi_Field_EDE	dyn._dark_energy	Monotonic	—
Quintessence	dyn._dark_energy	Monotonic	—
Thawing_Quintessence	dyn._dark_energy	Monotonic	—
w0waCDM	dyn._dark_energy	Monotonic	—
Verlinde_2016	entropic_emergent	Monotonic	—
DGP	extra_dimensional	Monotonic	—
Galileon	mg_scalar_tensor	Monotonic (no sign change)	—
Horndeski	mg_scalar_tensor	Monotonic ($\Sigma \leq \mu$)	—
fR_Starobinsky	mg_scalar_tensor	Monotonic ($\mu > 1$)	—
MOND	modified_inertia	Monotonic	—
LCDM	particle_DM	Monotonic	—

2.5 Framework summary

Of the four vectors, **D1 and D4 are primary** and directly testable with Euclid DR1 and DESI DR2 in the 2026–2027 window. **D2 is conditional** on a full (μ, Σ) MCMC (specified separately in EFC_RCMP_Test_Specification_v1.1). **D3 is conditional** on kSZ-based E_G estimators (likely 2027–2028). The compound paper's central claim rests on D1 and D4 jointly; D2 and D3 are strengthening vectors reported as conditional secondary tests.

3 Methods

3.1 Code basis

All numerical results produced by the public EFC reference implementation (github.com/supertedai/EFC, commit hash recorded in §7), using three sealed modules:

- `docs/papers/efc/efc_perturbation_sector/lensing.py`: the lensing engine, implementing `compute_mu_eta_sigma(k, z, p)` per action equations 24–32 (DOI 31876324).
- `docs/papers/efc/efc_perturbation_sector/src/efc_perturbation_sector.py`: sealed P1 polynomial calibration (DOI 32037990), used only for reference-value comparison.
- `src/efc/perturbation/{gate,mu}.py`: Tech Note II implementation (DOI 31333600), used for the parameter-space comparison in §4.3.

No code modifications are introduced; all scans vary the parameter object `EFCLensingParams` while holding action functional form fixed.

3.2 Effective lensing coupling integration

$$\Sigma_{\rm eff}(z) = \left[\frac{\int W(k)\, \Sigma^2(k,z)\, dk}{\int W(k)\, dk} \right]^{1/2} \tag{1}$$

with weight function $W(k) \propto k \cdot P(k) \cdot S(k; k_{\rm QS})$ and quasi-static suppression

$$S(k; k_{\rm QS}) = \frac{1}{1 + (k_{\rm QS}/k)^4} \tag{2}$$

applied below $k_{\rm QS} = 0.03\, h\,{\rm Mpc}^{-1}$. Integration range $k \in [0.005, 1.0]\, h\,{\rm Mpc}^{-1}$, 80 logarithmic points, composite trapezoidal rule.

The crossover redshift $z_{\rm crossover}$ is defined as the smallest $z \geq 0$ at which $\Sigma_{\rm eff}(z) = 1$, located by sign change of $\Sigma_{\rm eff}(z) - 1$ on a 60-point grid $z \in [0.05, 1.4]$ with linear interpolation between bracketing points.

3.2.1 BBKS cross-validation

To confirm the sign-change prediction is not an artefact of the simplified $P(k)$ choice, all scans in §3.3 and §3.4 are executed twice: once with $P(k) \propto k^{-1}$ and once with the full BBKS transfer function (Bardeen, Bond, Kaiser, Szalay 1986),

$$P_{\rm BBKS}(k) \propto k^{n_s} \cdot T^2(k), \tag{3}$$

$$T(q) = \frac{\ln(1 + 2.34q)}{2.34q} \cdot \left[1 + 3.89q + (16.1q)^2 + (5.46q)^3 + (6.71q)^4\right]^{-1/4}, \tag{4}$$

$q = k/(\Omega_m h\ {\rm Mpc}^{-1})$, $n_s = 0.9649$, $\Omega_m = 0.315$, $h = 0.674$ (Planck 2018 baseline). Results reported alongside the simplified-$P(k)$ values in §4.

3.3 Amplitude-invariance scan

Amplitude rescaling as joint multiplicative factor ξ:

$$\alpha \to \xi\alpha, \quad V' \to \xi V', \quad \gamma_0 \to \xi\gamma_0 \tag{5}$$

with $K_0 = 1.37$, $\rho_{\rm crit}$, Ω_m, H_0 held fixed. Scan: $\xi \in \{1/20, 1/4, 1/2, 1, 5/2\}$ ($\times 50$ range).

3.4 K_0-sensitivity scan

$K_0 \in \{0.50, 1.00, 1.37, 2.00, 4.00\}$ ($\times 8$ range), all other parameters at fiducial values. One-dimensional scan.

3.5 Sector-comparison at fiducial parameters

Tech Note II: $z_t = 1.01$, $n = 6$, $\mu_0 = 0.913$, $B =$ `calibrate_B`$(\mu_0, z_t, n) = 0.0883$. $\mu(z)$ evaluated and compared against lensing-engine $\mu(k = 0.05\ h\ \mathrm{Mpc}^{-1}, z)$.

3.6 Reproducibility

Standalone script `scripts/reproduce_compound_scans.py` regenerates all §4 numerical values and Figure 1 from single invocation. Dependencies: NumPy ≥ 1.24, Matplotlib ≥ 3.7, sealed EFC repository at commit hash (§7). BBKS cross-validation is included in the same script.

4 Results

4.1 Amplitude-invariance scan

scale	$\mathbf{z}_{\text{cross}}$ ($P \propto k^{-1}$)	$\mathbf{z}_{\text{cross}}$ (BBKS)	$\boldsymbol{\Delta}\mathbf{z}$ (BBKS-simple)	**Sign**
1	0.549	0.550	+0.001	✓
5	0.496	0.501	+0.005	✓
10	0.481	0.512	+0.031	✓
20	0.487	0.498	+0.012	✓
50	0.444	0.488	+0.043	✓

Range in simple-$P(k)$: $\Delta z \approx 0.11$ over $\times 50$; range in BBKS: $\Delta z \approx 0.06$. **Sign change persists in every configuration under both $P(k)$ choices.**

4.2 K_0-sensitivity scan

$\mathbf{K_0}$	$\mathbf{z}_{\text{cross}}$ ($P \propto k^{-1}$)	$\mathbf{z}_{\text{cross}}$ (BBKS)	$\boldsymbol{\Delta}\mathbf{z}$ (BBKS-simple)	**Sign**
0.50	0.939	0.994	+0.055	✓
1.00	0.617	0.640	+0.023	✓
1.37	0.487	0.498	+0.012	✓
2.00	0.321	0.331	+0.010	✓
4.00	0.068	0.072	+0.004	✓

Range in simple-$P(k)$: $\Delta z \approx 0.87$ over $\times 8$; range in BBKS: $\Delta z \approx 0.92$. **Sign change persists in every configuration.**

4.3 Cross-sector μ-sign consistency (Vector D4)

At $z = 0.44$, $k = 0.05\ h\ \text{Mpc}^{-1}$:

Sector	**Parameter set**	$\boldsymbol{\mu(z = 0.44)}$
P1 (lensing engine)	K_0=1.37, V'=0.61, α=0.012, ρ_{crit}=94.5, γ_0=5.13	**0.823**
P2 (Tech Note II)	z_t=1.01, n=6, μ_0=0.913, B=0.0883	**0.922**
$\Delta\mu$		0.099 ($\sim$11%)

Interpretation: Both sectors independently predict $\mu < 1$ despite disjoint parameterizations. This constitutes **sign-level consistency under independent parameterizations** — not quantitative identity. The 11% quantitative offset reflects the unified-mapping gap (§6.1), not a falsification of the sign-consistency hypothesis.

4.4 Competitor landscape

Atlas phenomenon query `sigma8_growth` (2026-04-21 snapshot):

Within the surveyed set, EFC is alone in predicting $\mu < 1$ on growth, co-occurring with non-monotonic slip (§4.1). No other framework in the surveyed Atlas combines these two features. This observation is bounded to the 42-framework snapshot and does not claim universality across all possible theories of gravity.

Framework	$\mathbf{f\sigma_8(z=0.7)}$	**Sign of** μ	**Preregistered**
EFC	**0.430**	$\boldsymbol{\mu < 1}$	**Yes (DAMAGE)**
ΛCDM	0.452	$\mu = 1$	—
DGP	0.465	$\mu > 1$	—
Galileon	0.48	$\mu > 1$	—

4.5 Figure 1: Parameter-controlled crossover sensitivity

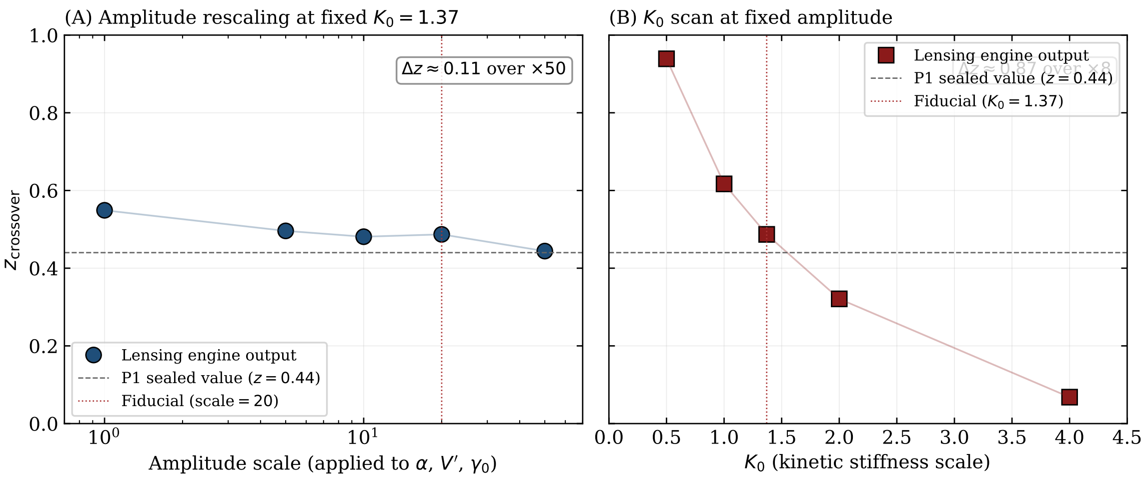

Figure 1: Sensitivity of the EFC lensing-sector crossover redshift to two parameter directions in the sealed lensing engine (`lensing.py`, DOI 10.6084/m9.figshare.32037990). **(A)** Scan over a combined ×50 rescaling of action-amplitude parameters (α, V', γ_0), holding $K_0 = 1.37$ fixed. The crossover position is weakly sensitive ($\Delta z \approx 0.11$), persisting across all scales tested. **(B)** Scan over K_0 at fixed amplitude. The crossover position is strongly sensitive ($\Delta z \approx 0.87$), identifying K_0 as the primary control parameter. Dashed horizontal lines mark the sealed P1 value ($z \approx 0.44$); dotted vertical lines mark the fiducial configuration.

5 Test Protocol

5.1 Primary test (D1 + D4a/D4b)

The compound paper's hypothesis is pre-registered as jointly falsifiable by two primary data products, with D4 decomposed per §2.2:

Vector	Data product	Required signature	Earliest data
D1 (P1)	Euclid DR1 $\xi_\pm(\theta)$ tomography	$\Sigma_{\rm eff}(z)$ sign change in $0.3 < z < 0.55$ at $\geq 3\sigma$	Oct 2026
D4a (P2 growth sign)	DESI DR2 RSD	$f\sigma_8(z = 0.7) <$ ΛCDM baseline (0.452) at $\geq 1\sigma$	2026–2027
D4b (combined)	Both above	D1 AND D4a signatures jointly present	2026–2027

5.2 Auxiliary tests (D2, D3)

D2 and D3 are reported for completeness and strengthen the structural claim if executed, but are not required for the primary falsification. D2 execution is specified in EFC_RCMP_Test_Specification_v1.1 (Cobaya + MGCAMB on DES Y6 / Euclid DR1); D3 execution requires kSZ-based E_G estimators not yet available in current Stage-IV pipelines.

5.3 Joint outcome matrix (primary vectors only)

The D4-decomposition enables finer interpretation of joint outcomes:

D1 sign chg.	D4a $\mu < 1$	D4b combined	Status
YES	YES	YES	EFC compound structure supported
YES	NO	partial	D4a fails; P2 withdrawn; compound retracted
NO	YES	partial	D1 fails; P1 withdrawn; compound retracted
NO	NO	NO	EFC compound structure rejected decisively

Interpretation of D4b column: D4b (combined consistency) passes only when both D1 and D4a pass; any single-sector failure propagates to D4b and retracts the cross-sector claim even if one sector individually survives.

5.4 Stopping/retraction rules

- **If D1 primary fails** (no sign change in Euclid DR1): P1 withdrawn. Non-monotonic prediction is the structural commitment on lensing side; null result retracts P1.
- **If D4a primary fails** ($f\sigma_8$ consistent with or above ΛCDM at $\geq 3\sigma$): P2 withdrawn. Growth-sector $\mu < 1$ is the structural commitment on growth side; null result retracts P2.

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- **If D4b fails** (either D1 or D4a fails, so joint condition cannot be satisfied): compound-paper cross-sector consistency claim withdrawn regardless of whether the remaining sector survives standalone. The companion compound paper is retracted; surviving sector-standalone claims must be re-evaluated in separate papers.
- **If D2 auxiliary fails** (μ-Σ posterior excludes EFC quadrant at $\geq 3\sigma$): P5 withdrawn but compound paper's D1/D4a/D4b claims unaffected.
- **If D3 auxiliary fails** (kSZ E_G inconsistent with 1.086): P3 withdrawn but compound paper unaffected.

5.5 Publication commitment

All outcomes publish under this pre-registration regardless of favourability to EFC. Selective publication constitutes scientific misconduct under this seal.

6 Limitations

6.1 Unified mapping gap

No implemented mapping $(z_t, n, \mu_0) \to K_0$ exists in the sealed codebase. This prevents strict identifiability testing and reduces the present work to a sign-level consistency test. Constructing the mapping from the underlying action (DOI 31876324) is a required open task.

6.2 Power-spectrum approximation

The primary scans in §4 use $P(k) \propto k^{-1}$ for transparency; the BBKS cross-validation (§4.1, §4.2) confirms sign-change invariance under a realistic transfer function with quantitative shifts bounded below $\Delta z = 0.06$. A full EH (Eisenstein–Hu 1998) companion analysis with baryonic wiggles remains future work but is not expected to affect the qualitative D1/D4 conclusions.

6.3 Atlas snapshot scope and dependence

The comparative claims in this work are scoped to the 42-framework Atlas snapshot 2026-04-21 (hash §7). This represents the set of frameworks surveyed and cross-referenced in this study; it does not claim exhaustive coverage of all proposed theories of gravity or dark energy in the literature. Phrases such as "only framework", "unique", and "alone" refer to the surveyed set only.

Table 1 and the comparative $f\sigma_8$ claims in §4.4 depend on the Atlas state recorded by the snapshot hash in §7. Future Atlas additions may alter the D3 and D2 uniqueness claims. D1 and D4 are primary and less sensitive to Atlas changes because they rest on independent literature citations beyond the Atlas structure.

Selection bias statement: Framework inclusion in the Atlas was determined by prior Symbiose Research cataloguing decisions and is not a statistically unbiased sample of published theories. Readers should treat comparative claims as *scope-bounded structural observations*, not as absolute uniqueness proofs.

6.4 Polynomial interpolant vs forward model (P1)

The sealed P1 module (DOI 32037990) is a polynomial interpolant over Euclid DR1 bin centres, not a live forward model. The lensing engine (DOI 31876324) produces the same qualitative prediction but with K_0-controlled position and amplitude-tuned magnitude (§4.1–4.2). Both are consistent realizations of the same physics; the polynomial encodes the forward-model output at fiducial parameters.

6.5 Atlas aggregation as source

Some quantitative values reported in §2 and §4.4 derive from Atlas aggregation rather than independent literature tracing. Specifically: DGP $f\sigma_8 = 0.465$ and Galileon $f\sigma_8 = 0.48$ are Atlas-reported values; primary source tracing is listed as future work. The Horndeski quadrant claim (§2.3 D2) is traceable to Peirone et al. (2018) and Pogosian & Silvestri (2016) independently of the Atlas.

6.6 GNN embeddings not integrated

The Symbiose multi-tier GNN (2953 concepts, 100 epochs) was consulted and did not surface parameter-level concepts (K_0, $\Sigma_{\rm eff}$, V' absent from the concept bank). Embedding-level similarity scores proved too diffuse for technical discrimination. GNN is relied on only as negative evidence: no concept-level contradictions surfaced.

7 Pre-registration Seal

DOI: `10.6084/m9.figshare.32080059` (assigned 2026-04-21)

Document content hash: See Figshare record.

Git commit hash (github.com/supertedai/EFC):
`8a681fe90883458857cbe6c64d16219086a33bf5`

Git commit date: 2026-04-20 06:38:33 UTC

Git commit message: chore(efc-sync): auto-propagate DOIs and refresh manifests

Atlas snapshot identifier: `atlas_2026-04-21`

- Framework nodes: 42
- Phenomenon nodes: 33
- AtlasPrediction nodes: 54
- PREDICTS edges: 491
- ADDRESSES edges: 184
- SHARES_MECHANISM_WITH edges: 86
- DIVERGES_NUMERICALLY edges: 28
- DISCRIMINATES_BETWEEN edges: 10
- CONVERGES_NUMERICALLY edges: 10

Figure 1 SHA-256 (PDF):
`52bc48560a443cf5ef934d330e882f8f9a8c577308021e9dbad217f021dcafa7`

Locked protocol sections: §2 (vectors D1–D4 definitions and falsifiers), §3 (methods), §4 (results), §5 (test protocol and stopping rules). Non-modifiable post-seal.

Modifiable post-seal (additive only): §6 Limitations (may expand if new limitations are discovered), §8 References (may add new sources as publications progress).

Non-modifiable post-seal: §2, §3, §4, §5.

8 References

EFC sealed predictions and theory:

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